

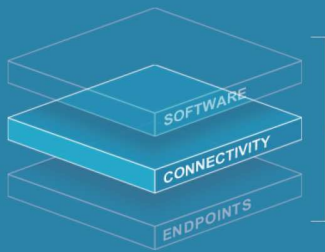
Indy RS500 RAIN RFID Reader Module

The Impinj Indy RS500 is a turnkey reader solution packaged as a surface-mount module. Indy RS500 makes embedding RAIN RFID quick and easy with low development risk and fast time-to-market.



Embedded Solution for Proximity Use Cases

Ideal for applications requiring proximity reads of small tag populations, Indy RS500 surface-mount modules enable a diverse range of connectivity devices such as printers, handheld readers, tablets, POS, access control terminals, smart kiosks, smart mirrors, and shelf readers.



The Impinj Platform includes Connectivity Devices



Indy RS500 Benefits

Turnkey Solution

Completely integrated and fully tested to meet worldwide compliance with module certification

Pin Compatibility with Indy RS1000

Enables a single hardware design for both higher-performance and lower-cost applications

Ideal for Proximity Readers

Delivers read reliability for small tag populations within short read range

Key Features

➤ Easy System Integration

IRI (Impinj Radio Interface) API, Indy Tool Kit and sample code simplify the design and integration process

➤ Typical Read Range

Optimal for applications requiring up to 3 meters read range

➤ Modular Certification

FCC and Canada certified and tested to meet all major regulatory regions worldwide

Use Cases

! Loss Prevention

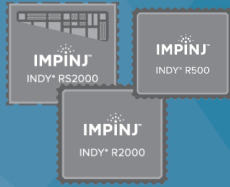
Ensure security and process control for people and assets with Indy RS500-enabled fixed portal readers

👍 Shopper Experience

Interactive mirrors and smart fitting rooms can identify products in proximity with Indy RS500 embedded readers

🏠 Inventory Accuracy

Drive sales with enhanced store, in-transit or warehouse inventory visibility with embedded readers based on Indy RS500



Reader Chip and Module Family Overview

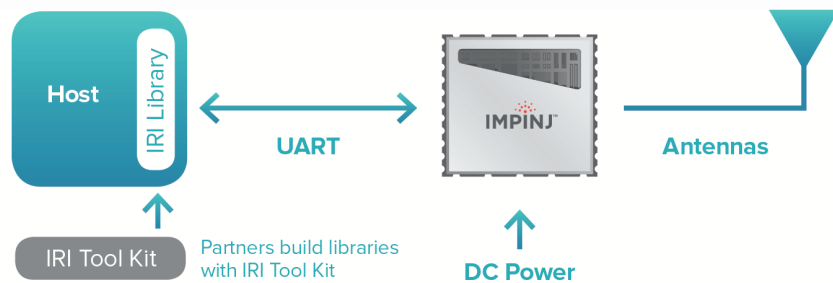
Indy reader chips and surface-mount modules provide the performance and flexibility needed to enable a full range of RAIN RFID solutions. Indy surface-mount modules provide low development risk and fast time-to-market while Indy chips offer a path to future cost optimization, all based on a common Indy platform.

Product Details	INDY RS500	INDY RS1000	INDY RS2000
Air Interface Protocol	RAIN RFID / ISO 18000-63 and EPCglobal Gen2v2 compliant		
Typical Read Range*	≤ 3 m	≤ 6 m	≤ 10 m
Output Power	10 to 23 dBm	10 to 27 dBm	10 to 31.5 dBm
Rx Sensitivity	-65 dBm	-75 dBm	-74 dBm
Antenna Ports	1		4
DC Power Supply	3.6 to 5.25 volts		3.2 to 5.25 volts
Power Consumption	2.5 watts @ 23 dBm	3.5 watts @ 27 dBm	7 watts @ 30 dBm
Temperature	Operating: -20 to +70 °C Storage: -30 to +100 °C		Operating: -20 to +60 °C Storage: -30 to +100 °C
Dimensions	29 x 32 mm		38 x 52 mm

* Assumes 6 dBi reader antenna and far-field tag with Impinj Monza R6 chip

THREE SIMPLE CONNECTIONS

1. DC Power
2. UART Communication
3. RF Antennas



Ready to discuss how Impinj can help your business?

Impinj (NASDAQ: PI) wirelessly connects billions of everyday items such as apparel, medical supplies, and automobile parts to consumer and business applications such as inventory management, patient safety, and asset tracking. The Impinj platform uses RAIN RFID to deliver timely information about these items to the digital world, thereby enabling the Internet of Things.

